

Nikhil Sunder

Curriculum Vitae

Coral Gables, Florida — nss106@miami.edu — Personal Site

EDUCATION & EMPLOYMENT HISTORY

B.S.B.A Quantitative Economics and Finance

Expected Completion: May 2027

Miami Herbert Business School, University of Miami

Minor in Math

Relevant Coursework: Calculus I-III, Linear Algebra, Differential Equations, Python Programming, Econometrics, Game Theory, Statistics, International Monetary Economics, Investments and Securities, Speculative Markets and Derivatives, Advanced Macroeconomic Theory, Mathematical Economics,

Research Fellow

Time Period: January 2026 - Present

Intelligent Computer Systems Research Institute (ICSRI),

Miami Herbert Business School, University of Miami

Department of Business Technology

Conducted independent research on neural spectral state-space models for non-stationary econometric forecasting, resulting in a peer-reviewed publication at FLAIRS-39 and a second manuscript currently under review. Designed and implemented the reference architecture, training pipelines, and benchmark suite for the Non-Stationary Spectral Decomposition Network (NS-SDN).

Undergraduate Researcher

Time Period: August 2025 - August 2026

Miami Herbert Business School, University of Miami

Department of Finance

Member of the Autonomous Fed research group, which studies reinforcement learning as a framework for monetary-policy rule discovery. Replicate and extend a Bundesbank study of RL-based policy agents, constructing simulated macroeconomic environments from time-varying-parameter structural VARs and non-stationary state-space models and training policy agents against them. Responsible for the econometric environment layer: model specification, estimation, and the data pipeline feeding the simulator, built on fedfred.

Research Assistant

Time Period: September 2026 - Present

Miami Herbert Business School, University of Miami

Department of Business Technology

Paid part-time researcher for the Autonomous Fed research group, responsible for the neural-network and source-code backends of the project's RL infrastructure. Design and implement the policy-agent training stack — network architectures, training loops, and environment interfaces — and own the software engineering of the shared codebase: package structure, typing and testing standards, CI, and reproducible builds, bringing the group's research code to release-grade quality consistent with maintaining fedfred and edgar-sec.

PUBLICATIONS

Reviewed Articles

Sunder, N. (2026). Non-Stationary Spectral Decomposition Network for Econometric Time Series Forecasting. The International FLAIRS Conference Proceedings, 39(1). <https://doi.org/10.32473/flairs.39.1.141588>

Sunder, N., Lean, M. (2026). Non-Stationary Spectral Decomposition Network: Adaptive Spectral Emission Heads and Frequency Modulation. IEEE ORSS Proceedings, <https://ieeexplore.ieee.org/document/11684282>

Software (Archived, Citable)

Sunder, N. (2025). fedfred: A Python client for the Federal Reserve Economic Database (FRED) API (Version v3.0.0) [Computer software]. Zenodo. <https://doi.org/10.5281/zenodo.17635942>

RESEARCH AND SOFTWARE PROJECTS

Research Projects

Non-Stationary Spectral Decomposition Network (NS-SDN)

Neural spectral state-space architecture for non-stationary econometric time series: learned frequency decomposition with adaptive spectral emission heads and frequency modulation. Produced one peer-reviewed publication (FLAIRS-39) and one manuscript under review; reference implementation released as the ns-sdn package.

Autonomous Fed: Reinforcement Learning for Monetary Policy

Replication and extension of a Bundesbank reinforcement-learning policy study. Constructs simulated macroeconomic environments from time-varying-parameter structural VARs and non-stationary state-space models and trains policy agents against them. Status: in progress.

Software Projects

fedfred: A modern python package for interacting with the Federal Reserve Bank of St. Louis FRED, GeoFRED, ALFRED, and FRASER APIs.

PyPI and conda-forge; approx. 150,000 downloads; OpenSSF Best Practices certified; Codecov-tracked test coverage; Zenodo DOI above.
repository — documentation

edgar-sec: A feature-rich python package for interacting with the US Securities and Exchange Commission API: EDGAR

Currently in redevelopment by open-source team.
repository — documentation

toros: A powerful DataFrame extension for representing complex financial objects.

Currently in new development phase as part of edgar-sec redevelopment.
repository

cultivars: A modern computing library in python for providing vector autoregressions and other autoregressive economic models.

Current active development project, v1.0.0 expected release September 2026
development branch

ns-sdn: A modern neural architecture in python for decomposing non-stationary time series into interpretable trend and spectral components.

Software component of Non-Stationary Decomposition Network papers. Awaiting final review of IEEE paper, prior to publishing v1.0.0
repository

knuckleball: A modern python package for auditable, health-tested entropy with exact replay for stochastic scientific computation. random draw analysis and entropy source health testing abstraction for cultivars, ns-sdn, and autofed. Current ICSRI 2026-2027 project. In proposal phase.

autofed: Pythonic reinforcement learning for monetary policy.

Software component for Autonomous Fed research group paper. Built as a surface layer on top of fedfred. In research phase.

SERVICE AND PROFESSIONAL DEVELOPMENT

Service to the Discipline and Profession

Maintainer of open-source scientific software used in academic and applied economics; issue triage, release management, and documentation for fedfred and edgar-sec. Primary developer and maintainer of fedfred, edgar-sec, cultivars, toros, ns-sdn, knuckleball, and autofed.

Professional Development

Presented at FLAIRS-39 (Florida Artificial Intelligence Research Society).

Presented at IEEE-ORSS 2026 (Institute of Electrical and Electronics Engineers - Opportunity Research Scholars

Symposium).